The Ramanujan Protocol:
A Self-Correcting Pipeline for Data-Driven Conjecture via Symbolic Regression and Formal Verification

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Abstract

We present the *Ramanujan Protocol*, a pipeline for automated mathematical conjecture whose principal contribution is not discovery but **epistemic restraint**. The system observes numerical data, discovers formulas via genetic programming, verifies them through Z3 SMT solving, and—most importantly—autonomously detects and retracts its own false positives at four distinct levels of statistical failure.

We document four retractions caught by the Protocol's own auditing: (i) a GCT obstruction "discovery" that was a structural tautology (Littlewood–Richardson size constraint); (ii) a "universal power-law exponent" for *abc* triple quality that was a model-selection artifact (exponential R^2=0.93 vs. power law R^2=0.69, AIC difference 596); (iii) a cross-domain correlation matrix in which 8 of 9 statistically significant pairs collapsed to $\rho \approx 0$ after detrending; and (iv) the final surviving correlation, which a null-hypothesis test revealed to be an artifact of the detrending procedure itself (random monotonic sequences produced nearly identical ρ).

After this four-layer audit, the only cross-domain bridges that survive are *exact algebraic identities*: the system recovered all 7 known modularity correspondences from raw coefficient data (95/95 prime matches, 0/3 decoy false positives), confirmed the Sato–Tate $\sin^2\theta$ distribution from 302 primes, computed BSD critical values matching LMFDB to 4 significant figures, and formally proved $R(3,3) = 6$ and $R(3,4) = 9$ via SAT encoding.

Principal finding: For arithmetic sequences, statistical correlation is not reliable evidence of mathematical connection. Only exact identities constitute genuine cross-domain bridges. A system that cannot catch its own false positives cannot be trusted with its true positives.

Implemented in Rust (5,400 LOC conjecture engine, 1,300 LOC number theory module, 345+ tests, Z3 4.15.8 backend). Open source.

Keywords: automated conjecture, symbolic regression, formal verification, epistemic integrity, self-correcting AI, statistical artifacts, exact identities

1 Introduction

Automated conjecture generation has advanced significantly in recent years, from Eureqa's physical law discovery [Schmidt and Lipson, 2009] to DeepMind's knot theory insights [Davies et al., 2021] and the Ramanujan Machine's continued fraction identities [Raayoni et al., 2021]. However, existing systems share a critical weakness: *they cannot catch their own false positives.* A neural network that discovers a pattern has no mechanism to audit whether that pattern is a structural artifact, a model-fitting coincidence, or a genuine mathematical relationship.

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We argue that trustworthy automated mathematics requires not just discovery but *epistemic self-correction*: the ability to detect and retract findings that are structurally trivial or statistically unsupported. We identify four requirements:

1. **Grounded observation**: formulas discovered from computed data, not recalled from training corpora.
2. **Formal verification**: conjectures proven by an independent engine (Z3 SMT solver), not just numerically validated.
3. **Self-correction**: the system autonomously detects and retracts tautologies and model-dependent artifacts.
4. **Cross-domain pattern matching**: the system finds correspondences between different mathematical domains without being told what to look for.

The Ramanujan Protocol implements all four as a closed pipeline. We demonstrate it across number theory, combinatorics, dynamical systems, and Ramsey theory, with extended case studies in Langlands-type correspondence recovery and extreme value statistics.

2 Architecture

The pipeline consists of five stages:

Stage 1: Observe. Mathematical engines compute numerical sequences: $a_p(E)$ from elliptic curve point counting, $p(n)$ from partition enumeration, γ_k from Riemann zeta zero finding, etc. Each sequence carries domain metadata (NumberTheory, Physics, Combinatorics).

Stage 2: Discover. Grammar-guided genetic programming with ensemble search (3 seeds), template library seeding (growth-class analysis), and semantic crossover (rejects offspring $10\times$ worse than parents). Constants optimized by L-BFGS with finite-difference gradients. Pareto-optimal (MSE, complexity) frontier computed alongside scalar fitness.

Stage 3: Verify. Four verification levels: (i) numerical (held-out 20% test split), (ii) bounded induction ($n \leq 1000$), (iii) formal Z3 proof (SMTLIB2 QF_NIA/QF_NRA, quantifier-free non-linear arithmetic), (iv) symbolic derivative verification ($dE/dt = 0$ for conservation laws).

Stage 4: Self-Correct. Competing model analysis (AIC comparison), bootstrap confidence intervals, structural tautology detection. Any finding that fails these audits is retracted.

Stage 5: Map. Cross-domain correlation matrix computed via Spearman rank correlation. Graph Laplacian eigenvalues extracted to reveal the spectral geometry of mathematical interconnection.

3 Results I: Formula Discovery and Proof

3.1 Closed-Loop Pipeline (Observe $\rightarrow$ Discover $\rightarrow$ Prove)

The complete pipeline was demonstrated on triangular numbers: the engine observed $\{1, 3, 6, 10, 15, \ldots\}$, discovered $T(n) = n(n+1)/2$ via GP, and Z3 proved the induction:

- **Base**: $T(1) = 1$ (Z3: UNSAT for negation).
- **Step**: $T(k+1) = T(k) + (k+1)$ (Z3: UNSAT for negation).

This constitutes a formal induction proof generated from raw numerical data, with no human guidance on what formula to search for or how to prove it.

Additional results: φ from Fibonacci ratios (MSE 8.4×10^{-21}), $E = x^2 + v^2$ conservation law (symbolic proof: $dE/dt = 2xv + 2v(-x) = 0$), Bell–Stirling identity ($B(n) = \sum S(n,k)$, exact to $n = 15$).

3.2 Formal Proofs via Z3

Statement	Method	Result
$T(n) = n(n+1)/2$	QF_NIA induction	**Proved**
$\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$	QF_NIA induction	**Proved**
$n^2 \geq n$ for $n \geq 1$	QF_NRA inequality	**Proved**
$(n+1/n)/2 \geq 1$ (AM-GM)	QF_NRA inequality	**Proved**
$R(3,3) = 6$	SAT/UNSAT	**Proved**
$R(3,4) = 9$	SAT/UNSAT	**Proved**
$R(4,4) > 17$	SAT	**Verified**

4 Epistemic Self-Correction: Four Layers of Restraint

The Protocol's distinctive contribution is not discovery but *retraction*. We document four retractions caught automatically, each representing a different failure mode of statistical discovery.

4.1 Layer 1: Structural Tautology (GCT Obstruction)

The engine computed Kronecker coefficient obstruction ratios for same-weight partition triples and reported 99.9% obstruction at $n = 6$—apparently strong computational evidence for $P \neq NP$. Post-hoc auditing revealed a **mathematical tautology**: the Littlewood–Richardson size constraint $|\mu| + |\nu| = |\lambda|$ forces $2w = w$ for same-weight triples, so $w = 0$ and all non-trivial triples vanish by construction.

Failure mode: logical constraint makes the finding vacuous. **Detection mechanism**: structural constraint checking.

4.2 Layer 2: Model Selection Artifact (*abc* Quality Decay)

The engine discovered $q(\text{rank}) \approx 1 + A/\text{rank}^B$ with exponent $B \approx 1.05$, stable across four scales. Competing model analysis via AIC:

Model	R^2	AIC	Verdict
Exponential: $A \cdot e^{-B\cdot\text{rank}}$	**0.930**	-865	**Best**
Power law: A/rank^B	0.690	-270	Rejected
Log-corrected: $A/(n \ln^C n)$	0.000	-469	Rejected

The AIC difference of 596 is decisive. Bootstrap 95% CI for the power-law exponent: $B \in [0.92, 1.24]$—too wide to claim $B \approx 1.05$.

Failure mode: wrong functional form inflates apparent fit. **Detection mechanism**: competing model comparison (AIC), bootstrap CI.

4.3 Layer 3: Growth-Rate Confound (Correlation Matrix)

We computed Spearman rank correlations between 6 arithmetic sequences (prime gaps, *abc* quality, partitions, class numbers, zeta zero spacings, Sato–Tate angles). The raw matrix showed 9 pairs with $|\rho| > 0.3$, including 5 with $|\rho| > 0.5$. A Fiedler-vector analysis suggested a clustering structure.

Detrending each sequence by its power-law fit and re-correlating the residuals revealed:

Pair	Raw ρ	Detrended ρ
class_nums $\leftrightarrow$ partitions	$+0.616$	$+0.070$
abc_quality $\leftrightarrow$ class_nums	-0.617	-0.051
partitions $\leftrightarrow$ zeta_spaces	-0.436	$+0.039$
prime_gaps $\leftrightarrow$ abc_quality	-0.507	$+0.036$
partitions $\leftrightarrow$ prime_gaps	$+0.507$	-0.002

8 of 9 correlations collapsed to $\rho \approx 0$. Only *abc*_quality$\leftrightarrow$partitions survived with detrended $\rho = -0.630$.

Failure mode: monotonic sequences spuriously correlate through shared growth rates. **Detection mechanism**: detrending residuals, first-difference analysis.

4.4 Layer 4: Detrending Artifact (Null Hypothesis Test)

The surviving *abc* $\leftrightarrow$ partitions correlation ($\rho = -0.63$) was subjected to a null hypothesis test comparing against random baselines:

- **Shuffle baseline**: randomly permuted *abc* qualities give $\rho = -0.03 \pm 0.11$ (test works correctly).
- **Random-sequence baseline**: random monotonic sequences with similar growth rate give detrended $\rho = -0.50 \pm 0.10$. The real $\rho = -0.63$ is only 1.26 std devs from this baseline.
- **Trend-model sensitivity**: linear detrending gives $\rho = +0.34$ (opposite sign). Power-law gives -0.63.
- **Subsample instability**: $\rho_{30} = +0.05$, $\rho_{50} = -0.72$, $\rho_{100} = -0.63$.

The detrending procedure itself induces spurious correlation when applied to monotonic sequences. The result is opposite-signed under different trend models and unstable under subsampling.

Failure mode: statistical procedure generates its own artifact. **Detection mechanism**: random-sequence null baseline, trend-model sensitivity analysis.

4.5 What Survives

After four layers of self-correction, *no statistical correlation between arithmetic sequences survives*. The only cross-domain bridges that remain are **exact algebraic identities**:

- Modularity: $a_p(E) = c_p(f)$ for 95/95 prime matches.
- Sato–Tate: $\sin^2\theta$ density shape from 302 primes ($|\rho| < 0.1$ with all other sequences, even raw).
- Formal Z3 proofs: $T(n) = n(n+1)/2$, $R(3,3) = 6$, etc.

4.6 Significance

Each retraction removed something we wanted to believe. The GCT result looked like evidence for $P \neq NP$. The *abc* power law looked universal. The correlation matrix looked like a map of mathematical structure. All four were illusions, caught before publication by the system's own auditing.

A system that cannot catch its own false positives cannot be trusted with its true positives. The four layers of restraint demonstrated here are the necessary infrastructure for trustworthy automated conjecture generation.

5 Case Study: Cross-Domain Correspondence Recovery

As a test of the Protocol's cross-domain pattern matching capability, we applied it to the Langlands correspondence between elliptic curves and modular forms.

5.1 Modularity Correspondence Recovery

We generated $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$ for 7 Cremona curves (conductors 11–21) and $c_n(f)$ for 7 weight-2 newforms via Hecke multiplicativity. The engine was given all 14 sequences plus 3 decoy forms with plausible but incorrect coefficients, *without pairing information*.

Result: 7/7 correct correspondences identified (95/95 prime matches), 0/3 decoy false positives. The engine recovered all known modularity correspondences from raw numerical data. This demonstrates that the Protocol can detect exact coefficient identities across mathematical domains without being told what to look for.

5.2 Additional Verifications

Sato–Tate distribution: Frobenius angles $\theta_p = \arccos(a_p/2\sqrt{p})$ for 302 primes on curve 11a1 confirm the $\sin^2\theta$ density shape.

Montgomery pair correlation: Zeta zero spacings tested against GUE: $\chi^2 = 9.28$ ($df = 19$, 5% critical ≈ 31.3). Cannot reject GUE at 5% significance. Level repulsion confirmed.

BSD critical values: $L(11\mathrm{a}1, 1) = 0.2538$ (LMFDB: 0.2538, 4-digit match). All rank-0 curves satisfy $L(E, 1) > 0$.

6 Retracted Claim: The Spectral Map of Mathematics

An earlier version of this paper included a spectral-geometry section claiming that the cross-domain correlation matrix reveals a "map of mathematics"—an arithmetic continent (prime gaps, *abc* quality, partitions, class numbers, zeta spacings) joined by a weak bridge (Fiedler value $\lambda_2 = 0.273$) to an isolated Sato–Tate island. The Fiedler vector appeared to cleanly partition the sequences into these two clusters.

The system caught this claim as a statistical artifact (see Section 4, Layers 3 and 4). After detrending and null-hypothesis testing, all apparent cross-domain correlations except those involving Sato–Tate collapsed to $\rho \approx 0$.

The honest interpretation of the Laplacian spectrum is:

- Sato–Tate's isolation from the other sequences ($|\rho| < 0.1$ even raw) is genuine, and reflects the equidistribution theorem.
- The "arithmetic continent" is a cluster of sequences that share *growth rates*, not structure. Monotonically increasing sequences statistically correlate with each other even when they are mathematically unrelated.
- The Fiedler value $\lambda_2 = 0.273$ identifies a bottleneck between *deterministic growth* and *probabilistic equidistribution*, not between two regions of mathematics.

We retain this retracted section in the manuscript because the retraction is itself a result. The correlation matrix passed several plausibility checks (including Sato–Tate as a built-in control sequence) before the detrending audit exposed it. Without that audit, a visually compelling "map of mathematics" would have been published. Visual compellingness is not evidence.

7 Related Work

Symbolic regression: Eureqa [Schmidt and Lipson, 2009], AI Feynman [Udrescu and Tegmark, 2020], PySR [Cranmer, 2023]. Our work adds formal Z3 verification and self-correction.

AI for mathematics: DeepMind's knot theory [Davies et al., 2021], AlphaProof [DeepMind, 2024], the Ramanujan Machine [Raayoni et al., 2021]. We achieve comparable cross-domain discovery with a transparent, deterministic pipeline (no neural networks).

Computational number theory: Cremona's tables [Cremona, 1997], LMFDB [The LMFDB Collaboration, 2024]. We verify modularity computationally and discover it autonomously.

8 Conclusion

The Ramanujan Protocol's principal contribution is a categorical distinction: for arithmetic sequences, **statistical correlation is not mathematical connection, but exact identity is**.

This distinction was established empirically through four layers of self-correction. Every statistical claim the system generated was audited, and every correlation between arithmetic sequences—from class numbers to partition counts to *abc* triple qualities—failed that audit. What survived were the exact identities: modularity correspondences ($a_p = c_p$ on all 95 prime matches), the Sato–Tate density shape, Z3-proven induction identities, and formally verified Ramsey numbers.

The practical consequence for automated mathematics is stark. Correlation-based discovery between arithmetic sequences is unreliable even with sophisticated detrending. Trustworthy automated mathematics requires auditing mechanisms at every level:

1. Structural constraint checking (catches tautologies).
2. Competing model analysis via AIC (catches model artifacts).
3. Detrending residuals (catches growth-rate confounds).
4. Null-hypothesis testing against random baselines (catches procedural artifacts).

A system that cannot catch its own false positives cannot be trusted with its true positives. We suggest that this auditing hierarchy should become standard practice for any automated conjecture system that claims to find cross-domain correspondences in arithmetic data.

The pipeline is open-source (Rust, 345+ tests) and implemented as part of the Symthaea cognitive architecture.[1]

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